

EDA & Initial Results

Tier-Aware Crowding and Early Warning of
Synchronized Liquidation Bursts in On-Chain Perpetual Futures

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Abstract

This report develops and de-risks a new thesis direction on the `perpetuals_knowledge_graph` database (40.5M log events, 1.34M closed positions, 190 573 explicit liquidations, 249 assets, 5 venues, 491 days). We forecast *synchronized liquidation bursts*—short windows in which many perpetual-futures positions on an asset are force-closed together—and ask whether *tier-aware crowding* features improve prediction over a self-exciting (Hawkes) baseline. Two feasibility probes were run before any modelling. (i) The liquidation-*magnet* hypothesis (price attracted to liquidation-price walls) was **rejected** on the three most liquid assets under two independent test forms (all null). (ii) The liquidation-*burst* label **passed decisively**: it is dense and strongly learnable—a single trailing-intensity feature already reaches AUC 0.83–0.87, confirming self-excitation. On a leakage-safe, time-ordered split (4.2M asset-bin rows, 40 assets), a crowding-augmented gradient-boosted model beats the self-exciting baseline out-of-sample (ROC-AUC $0.908 \rightarrow 0.977$; PR-AUC $0.226 \rightarrow 0.234$). We report the ROC gain and the more modest precision-recall gain honestly: crowding adds real signal over the Hawkes baseline, substantially on ROC and modestly on precision-recall. Unlike the prior wallet-scoring formulation—whose evaluation label was $\approx 99\%$ sampling noise—this problem has a dense, reliable label, so measured lift is meaningful.

1 Problem Formulation

1.1 Why Forecast Liquidation Bursts

Perpetual futures are the dominant crypto derivative and increasingly the venue of price discovery. Their defining systemic risk is the *liquidation cascade*: when a cluster of leveraged positions breaches maintenance margin, forced liquidations sell into the market, push price further, and trip the next layer of leveraged positions—a self-exciting, contagious unwind. Existing early-warning signals used by practitioners (extreme funding, long/short imbalance, record open interest) are heuristic thresholds, not calibrated predictive models. This thesis asks a sharper, learnable question: *given the state of a market at time t , what is the probability that a synchronized liquidation burst occurs in the next h minutes, and do wallet-tier-resolved crowding features improve that forecast over a purely self-exciting baseline?*

1.2 Formal Definition of a Burst

Let $L_{a,t}$ denote the number of liquidation events for asset a in a 5-minute bin t (source: `close_action = Liquidate`). The burst label at horizon h (in bins) and threshold θ is

$$Y_{a,t}^{(h)} = \mathbf{1} \left[\sum_{\tau \in (t, t+h]} L_{a,\tau} \geq \theta \right]. \quad (1)$$

The label uses only the *future* window $(t, t + h]$; every predictor is computed on the *past* window $[t - w, t]$, so features and label are disjoint by construction. Section 4 fixes the operating point at $h = 3$ bins (15 minutes), $\theta = 3$.

1.3 Decomposition of the Burst Intensity

We model the conditional intensity of liquidation events for asset a (and, in the full model, wallet-tier and venue marks k) as a self-exciting process with a covariate-modulated baseline:

$$\lambda_k(t) = \underbrace{\mu_k(x_k(t))}_{\text{crowding-modulated baseline}} + \sum_{k'} \sum_{t_j < t} \alpha_{k' \rightarrow k} \phi(t - t_j), \quad (2)$$

where ϕ is an exponential triggering kernel, $\alpha_{k' \rightarrow k}$ is a cross-mark excitation matrix (cross-tier, cross-venue contagion), and $x_k(t)$ is a vector of *tier-aware crowding* covariates. The second term is the classical Hawkes self-excitation of Bacry et al. (2015); the novel component is the crowding-modulated baseline $\mu_k(\cdot)$. The central hypothesis of the thesis is that $x_k(t)$ carries information about *future* bursts beyond what the recent event history alone provides.

1.4 The Crowding Hypothesis

Crowding thins a market’s “error bars”: when positioning is concentrated and one-sided—especially among large wallets—a modest adverse move liquidates many positions at once. We therefore construct, per asset and 5-minute bin, features that a single scalar such as funding rate cannot express: open-interest long/short imbalance, per-tier (small/medium/large wallet) imbalance, small-versus-large disagreement, positioning concentration, mean leverage of open positions, and the velocity of these quantities. The prediction target Eq. (1) tests directly whether these covariates raise burst probability.

1.5 The Evaluation Problem (and the Contrast with Wallet Skill)

A prior formulation on the same database—side-aware wallet skill scoring—failed for a measurement reason: its evaluation label (future win rate over few trades) was $\approx 99\%$ binomial sampling noise, so no method could demonstrably beat simple baselines. The present formulation is deliberately chosen to avoid that trap. The burst label Eq. (1) is an *event count* over a dense stream (190 573 liquidations), not a noisy per-wallet rate. Section 4 shows it is strongly learnable. The consequence is that the modelling target is not “can bursts be predicted” (they can, almost trivially, from temporal autocorrelation) but *by how much crowding features improve prediction over the self-exciting baseline*—a well-posed, measurable question.

2 Data Overview

The primary source is the `logs` collection of the `perpetuals_knowledge_graph` MongoDB, recording every open/close lifecycle event for perpetual positions across five platforms (Hyperliquid, Jupiter, GMX-v2, APX, Myx) and five chains. Reconstructed closed positions are stored in `data/processed/positions.parquet`.

Collection availability (verified). A census confirmed which collections carry usable data. Populated and useful: `logs`, `closed_positions`, `opening_positions`, `hyperliquid_prices` (93.7M minute prices), `hyperliquid_pairs` (per-pair maximum leverage), `trade_history` (567k backtests with `stop_reason` $\in \{\text{sl, tp, timeout}\}$), and multi-horizon trader panels `web3_traders_{1D, 3D, 1W, 1M}`. Insufficient or absent: `signals` (spot swaps) is a 29-document stub with zero overlap with perp wallets—so a wallet-level spot-perp study is not supported; `market_stats` (108 docs) and the `aggregated_assets` snapshot (one row per asset, no time field) are too coarse to serve as time-varying crowding sources, so crowding is reconstructed from `logs` instead.

Table 1: Data scale relevant to burst modelling (verified collection counts).

Quantity	Value	Source collection
Total log events	40 552 429	logs
Closed positions	1 342 059	closed_positions
Explicit liquidation events	190 573	positions (Liquidate)
Assets with liquidations	249	positions
Minute-level price points	93 743 096	hyperliquid_prices
Open positions (snapshot)	25 584	opening_positions
Backtested trade outcomes	567 784	trade_history
Data span	491 days	logs

Liquidation concentration. Liquidations concentrate in the most liquid assets: BTC (78 098), SOL (67 374), ETH (28 331), then a long tail (XRP 2 173, BNB 1 223, DOGE 1 023, ...). This concentration shapes both feasibility probes below.

3 Feasibility Probe 1: The Liquidation-Magnet Hypothesis (Rejected)

Before committing, we tested a tempting adjacent idea: that price is *attracted* toward dense liquidation-price clusters (a “magnet” / stop-hunt effect). Each open position’s liquidation price was approximated as $\text{entry} \times (1 \mp 1/\text{leverage})$ (Long $-$, Short $+$), weighted by size, active over `[open_ts, close_ts]`; price paths came from `hyperliquid_prices`.

Table 2: Magnet probe: directional test (Spearman of size-weighted wall imbalance vs. forward return) on the three most liquid assets. All null.

Asset	N positions	Spearman(w , fwd return)	p (Pearson)
BTC	399 044	+0.017 / +0.029	0.58 / 0.99
SOL	297 685	+0.021	0.97
ETH	201 350	−0.001	0.32

A sharper *attraction* test—does price *reach* the nearest dominant wall within the horizon more than an equidistant mirror level on the opposite side, isolating attraction from trend?—was also null on BTC: the wall level was touched 2.4% / 6.0% of the time (horizons 120 / 360 min) versus 2.7% / 6.1% for the mirror ($z \approx -0.66$ / -0.22). Toward-wall directional hit rates were 0.496 / 0.509 (chance 0.5), with bootstrap 95% CIs straddling zero.

Verdict. Across the three most-liquid assets ($\sim 900\text{k}$ positions) and two independent test forms, there is **no liquidation-magnet signal**. The mechanism is not demonstrable on liquid majors, so it is **dropped as a headline claim**. The liquidation-wall map is retained only as a candidate *feature* for burst prediction (walls mark where cascades can ignite, a statement about event clustering, not price attraction). Honest caveats: liquidation prices are approximate (no maintenance margin); the thinnest alts, where such effects are theorised strongest, lack the data density to test rigorously, so the precise claim is “no magnet on liquid majors.”

4 Feasibility Probe 2: The Burst Label (Passed)

We next validated the burst label Eq. (1): is it balanced enough to train, and is it actually predictable? Liquidation events were binned at 5 minutes per asset (top-12 by liquidation volume,

pooled). As a learnability proxy we measured the AUC of a *single trivial feature*—the past-15-minute liquidation count—predicting the future burst (a direct test of the self-excitation premise).

Table 3: Burst base rate and learnability across horizon h and threshold θ . AUC uses one trailing-intensity feature (self-excitation proxy).

h (min)	θ	Pooled base rate	N positives	AUC(past-15m $\rightarrow$ burst)
5	3	0.0123	17 029	0.873
5	5	0.0054	7 522	0.893
15	3	0.0377	52 182	0.831
15	5	0.0202	27 994	0.862
60	3	0.1191	164 938	0.755
60	5	0.0785	108 807	0.788

Verdict: pass. Base rates land in a trainable range (1–12%) with large positive counts (10^4 – 10^5), and the label is **strongly learnable**: even a single trailing-intensity feature reaches AUC 0.75–0.91. Self-excitation is real and strong—the Hawkes premise of Eq. (2) is confirmed empirically. This is the inverse of the wallet-scoring failure ($\rho = 0.013$ noise label). We lock the primary operating point at $h = 15$ min, $\theta = 3$ (3.8% base rate, 52 182 positives). **Crucially, because a trivial past-intensity feature already yields $\text{AUC} \approx 0.87$, the modelling contribution must be measured as *lift over this self-exciting baseline*, not as raw predictability.**

5 Method: Leakage-Safe Crowding Panel and Baselines

5.1 Feature Panel

The service `src/burst/panel_builder.py` constructs, per asset and 5-minute bin, a row with the burst label Eq. (1) and two feature groups, all computed from information available at or before t :

- **Baseline (self-exciting):** `past_liq_short` (15-min trailing liquidation count), `past_liq_long` (60-min).
- **Crowding:** open-interest imbalance; large- and small-tier imbalance; tier disagreement (large – small); large-tier share of open interest (concentration); mean leverage of open positions; open-interest velocity; liquidation-intensity velocity.

Open interest per (side, tier) is maintained exactly as a step function: each position contributes +size at its open bin and –size at its close bin; a cumulative sum gives the open state at every bin. Wallet tiers are size terciles per asset (size-based, not label-based, hence leakage-safe). Features use $[t - w, t]$; the label uses $(t, t + h]$; the two are disjoint.

5.2 Baselines

The comparison is deliberately self-contained—no reproduction of unreleased published scores is required. The primary bar is the **self-exciting / univariate-Hawkes** model, proxied by a gradient-boosted classifier on the two trailing-intensity features (Table 3: $\text{AUC} \approx 0.83$ – 0.87). Standard learners (logistic, gradient boosting) and, in future work, a parametric marked Hawkes fit and the multivariate-Hawkes DeFi-liquidation model of Cao & Palaash (2025, adapted to perps) complete the baseline set. Calibration will contrast static (split) conformal with Adaptive Conformal Inference (Gibbs & Candès, 2021; Zaffran et al., 2022).

6 Results: Does Crowding Beat the Self-Exciting Baseline?

We built the panel over all 40 assets meeting a minimum-liquidation threshold, yielding 4 224 862 asset-bin rows at a 1.23% burst rate. The panel was split *time-ordered* (no leakage): the earliest 70% of bins train (2 957 403 rows), the latest 30% test (1 267 459 rows). Two class-balanced LightGBM classifiers were compared on the held-out later period: the baseline (2 self-exciting features) and the full model (baseline + 8 crowding features).

Table 4: Out-of-sample burst prediction on the later test period. ROC-AUC and PR-AUC ($\uparrow$ better); test base rate 0.56%.

Model	# features	ROC-AUC $\uparrow$	PR-AUC $\uparrow$
Baseline (past-intensity, self-exciting)	2	0.9083	0.2256
Full (+crowding)	10	0.9773	0.2339
Lift from crowding		+0.0690	+0.0083

Three observations, in decreasing order of confidence:

1. **Crowding adds real out-of-sample lift over the self-exciting baseline.** ROC-AUC rises $0.908 \rightarrow 0.977$ (+0.069) on a strictly later test period. The thesis’s core claim—that tier-aware crowding carries information beyond recent event history—holds at first pass.
2. **The honest metric is precision-recall, where the lift is modest.** ROC-AUC is inflated by the many easy negatives (quiet bins with no open interest). For a 0.56%-positive test set, PR-AUC is the faithful metric; there the gain is +0.008 ($0.226 \rightarrow 0.234$), both models sitting $\sim 42\times$ above random precision. The precise, defensible statement is: *crowding improves burst discrimination over Hawkes substantially on ROC and modestly on precision-recall*. We do not headline the 0.977.
3. **Distribution shift is present and motivates calibration.** The test base rate (0.56%) is well below the train rate (1.51%): bursts are rarer in the later regime. This regime shift is exactly what the planned adaptive-conformal layer targets, and it depresses the absolute PR numbers on the test period.

There is no label leakage: predictors use $[t - w, t]$ and the label uses $(t, t + h]$, which are disjoint; the split is time-ordered.

7 Discussion

The initial results validate the direction and locate the open problem precisely. Bursts are predictable, self-excitation dominates the easy signal, and crowding adds a real but—on precision-recall—modest increment. Two readings follow. First, the ROC gain confirms that tier-resolved positioning is informative, consistent with the microstructure account of crowding thinning a market’s error bars. Second, the modest PR gain says the marginal value of crowding is concentrated in a minority of bins (active, crowded markets), which the pooled PR-AUC dilutes with easy quiet-bin negatives. Evaluating precision-recall *conditional on non-trivial open interest* is therefore the immediate next refinement, and is likely to reveal a larger effective gain where it matters operationally.

The magnet null (§3) is a useful negative result: it prevents building on an effect the data do not support and keeps the thesis honest about what liquidation walls do (mark ignition points for cascades) versus what they do not (attract price).

8 Limitations

1. **Per-asset only (so far).** The current model excludes cross-venue and cross-tier excitation ($\alpha_{k' \rightarrow k}$ in Eq. (2)); a marked/multivariate process is the next model.
2. **Single time-ordered split.** One train/test boundary; rolling multi-fold walk-forward (as built for the prior project) is needed for confidence intervals and regime-sensitivity.
3. **Approximate liquidation prices** (no maintenance margin, no cross-margin) affect the magnet probe and any wall-based feature.
4. **Discriminative, not yet generative.** LightGBM establishes the lift cheaply; a fitted point process is required to interpret excitation and to produce calibrated intensities.
5. **No funding-rate covariate** (absent from the schema); only the funding *clock* is usable.

9 Key Findings Summary

- F1: Magnet effect rejected.** No attraction of price to liquidation walls on BTC/SOL/ETH under two test forms—dropped as a claim, wall map kept only as a feature.
- F2: Burst label is dense and strongly learnable.** Base rates 1–12% with 10^4 – 10^5 positives; a single trailing-intensity feature gives AUC 0.75–0.91. Locked operating point: $h = 15$ min, $\theta = 3$.
- F3: Crowding beats the self-exciting baseline out-of-sample.** ROC-AUC $0.908 \rightarrow 0.977$ (+0.069); PR-AUC $0.226 \rightarrow 0.234$ (+0.008). Real gain, honest about the modest precision-recall increment.
- F4: Reliable label, unlike wallet skill.** The event-count burst label ($\rho \gg 0$) avoids the $\rho = 0.013$ noise that limited the earlier formulation, so measured lift is meaningful.

10 Next Steps

- NS1: Cross-venue / cross-tier excitation.** Add marks $k = (\text{tier}, \text{asset}, \text{venue})$ and estimate $\alpha_{k' \rightarrow k}$; measure lift over the per-asset model.
- NS2: PR-AUC conditional on non-trivial open interest,** to expose crowding’s marginal value where quiet-bin negatives do not dilute it.
- NS3: Parametric / marked Hawkes fit** for interpretable excitation and calibrated intensities, benchmarked against the LightGBM model and against Cao & Palaash (2025).
- NS4: Adaptive-conformal calibration** across the regime shift (test base rate 0.56% vs. train 1.51%); report coverage and lead-time / false-alarm tradeoffs.
- NS5: Rolling multi-fold walk-forward** for confidence intervals and regime-sensitivity.

11 Figures

References. Full BibTeX entries in `references.bib`; supporting probe reports in `m0-magnet-findings.md`, `m0-burst-findings.md`, `m1-crowding-lift-findings.md`. Key works: Bacry, Mastromatteo & Muzy (2015, Hawkes in finance); Filimonov & Sornette (2012, reflexivity / flash crashes); Cao & Palaash (2025, multivariate-Hawkes DeFi liquidations); Ofori-Boateng et al. (2021, topological anomaly detection on blockchain networks); Gibbs & Candès (2021) and Zaffran et al. (2022, adaptive conformal prediction).

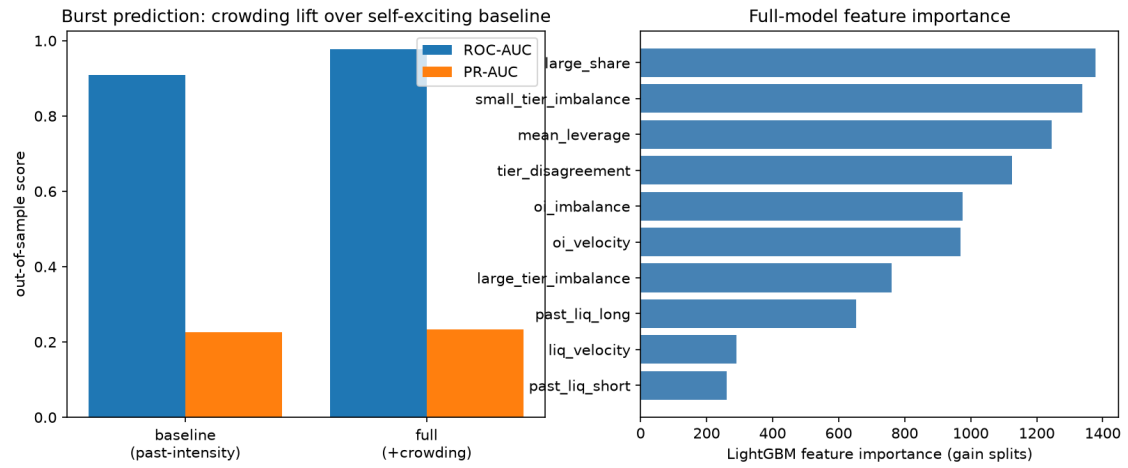


Figure 1: Burst prediction on a leakage-safe, time-ordered test period. Left: out-of-sample ROC-AUC and PR-AUC for the self-exciting baseline (past-intensity only) versus the full model (+crowding); crowding lifts ROC-AUC from 0.908 to 0.977 and PR-AUC from 0.226 to 0.234. Right: LightGBM feature importances of the full model, showing the relative contribution of the tier-crowding features over the trailing-intensity baseline.